

- 4.1 Here, the goal is to find the rising and falling propagation delays for this gate when it drives the input to h identical gates. Here, I'm assuming that the output of the gate under consideration drives *one* input to a second 2NOR gate, since all the fanout gates will be two-input gates (that said, each input consists of a pMOS transistor and an nMOS transistor, with separate gate capacitances).

Propagation delay t_{pd} refers specifically to the worst-case delay between a gate's input and output changing. Due to the physical characteristics of CMOS networks, this worst-case delay is often different depending on whether the gate is pulling up or down, and so we differentiate between input and output propagation delay. To account for the worst case pMOS series resistance in the pull-up network of the 2NOR gate, the pMOS gate width must be adjusted to be four times the size of the nMOS gate width to achieve the same worst-case equivalent rising and falling resistances of a unit inverter ($R = R$). See Fig. 1 for the drawn diagram of the 2NOR.

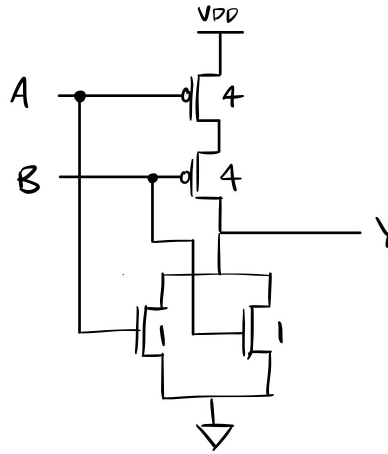


Figure 1: 2NOR inverter with equivalent resistance to a unit inverter.

To derive the worst-case rising and falling delay as a function of h identical gates being driven by this one, we can first find the equivalent resistance and capacitance of each component by using the RC delay model, and then, using the Elmore delay model, we can roughly calculate the effective delay as a function of h using our RC values.

In the pull-up network, each pMOS transistor will have a gate and diffusion capacitance of $4C$ and an equivalent channel resistance of $\frac{R}{2}$. In the pull-down network, each nMOS transistor will have a gate and diffusion capacitance of C and an equivalent channel resistance of R .

Beginning with worst-case falling delay t_{pdf} , we consider worst-case resistance on the path between Y and GND to be the case when a single nMOS transistor is on, and its equivalent resistance is R . Considering diffusion capacitances, the pMOS drain diffusion capacitance $4C$ is connected to Y, as well as two parallel nMOS drain diffusion capacitances $C + C = 2C$. We may ignore any pMOS diffusion capacitances not on the path between Y and GND, as well as the two nMOS source diffusion capacitances, as these are shorted to GND through the substrate tap. In total, diffusion capacitance sums to $4C + 2C = 6C$ for the driving 2NOR gate. On the other end, the gate capacitances of the pMOS and nMOS transistor at the load 2NOR gate sum to $4C + C = 5C$, and increase with each added fanout gate.

Using the formula for Elmore delay,

$$t_{pdf} = \sum_i R_{is} C_i = R(6C + 5hC) = (6C + 5h)RC,$$

Worst-case rising delay t_{pdr} occurs when both pMOS series-connected pMOS transistors are on. In this case we have two nodes to deal with, as opposed to single node in the falling transition. For the first node, at the junction between the source/drain of the series pMOS transistors, there is an equivalent resistance of $\frac{R}{2}$ from this node to VDD, and a diffusion capacitance of $4C$, assuming a fully-contacted but shared diffusion node between the transistors.

At the second node, corresponding to the output voltage, the equivalent resistance is comprised of both pMOS channels, so $\frac{R}{2} + \frac{R}{2}$, and the total diffusion and gate capacitance at this node consists of $6C + 5hC$, with h once again representing the number of load 2NOR gates. Summing the resistance and capacitance at the two nodes using the Elmore delay formula,

$$t_{pdr} = \sum_i R_{is} C_i = \frac{R}{2}(4C) + R(6C + 5hC) = (8 + 5h)RC.$$

4.2 The following (Fig. 2) is the stick diagram for a 2NOR gate, optimized to reduce delay due to parasitic capacitances:

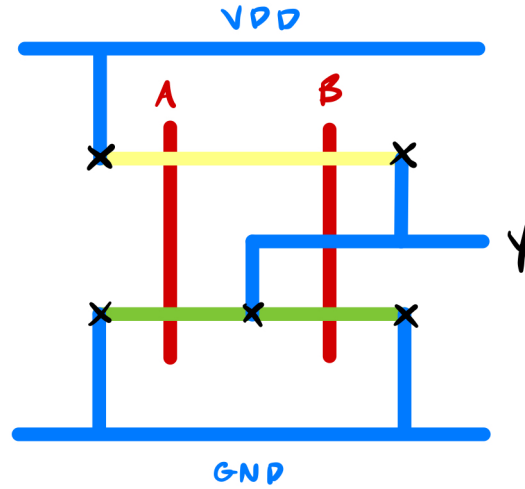


Figure 2: 2NOR stick diagram

Starting with rising delay, we assume that the shared diffusion node between the series pMOS inverters is not contacted and that the contacted diffusion node at the drains of the two parallel nMOS transistors is shared, we estimate a diffusion capacitance of $2C$ at node 1 and a total gate/diffusion capacitance of $5C + 5hC$ at the output node, with no change in equivalent resistance,

$$t_{pdr} = \frac{R}{2}(2C) + R(5C + 5hC) = RC(6 + 5h).$$

For falling delay, we again assume a shared, contacted diffusion between the two parallel nMOS drains, so

$$t_{pdf} = R(5C + 5hC) = RC(5 + 5h).$$

4.3 For an AOIXX gate, the Boolean logic function for n inputs to the first AND gate and m inputs to the second gate is

$$Y = \overline{(A_1 A_2 \dots A_n) + (B_1 B_2 \dots B_m)}.$$

In practice, this represents a pull-up network consisting of n parallel pMOS transistors in series with m pMOS transistors and a pull-down network consisting of n series nMOS transistors in parallel with m series nMOS transistors.

Targeting symmetric equivalent resistances in the worst case, we will size pMOS transistors at a width of 8λ , and the nMOS transistors at a width of λn for the "A" path to ground and $2\lambda m$ for the "B" path.

Neglecting gate capacitance, since the given AOIXX gate is not loaded, we'll start by deriving rising delay, t_{pdf} using the Elmore delay approximation. I will assume uncontacted, shared diffusion nodes for series connected transistors and contacted but shared diffusion nodes for parallel connected transistors. Ignoring capacitances that are shorted to VDD or GND by substrate/well taps, the worst case scenario is a series resistance of R through two pMOS transistor channels, with any parallel transistors lowering the equivalent resistance. At node 1, between the two networks of parallel pMOS transistors, the path to VDD will consist of a resistance of $\frac{R}{2}$ and a diffusion capacitance of